INCH-POUND

MIL-C-27072/89B
23 March 1993
SUPERSEDING
MIL-C-27072/89A
24 September 1987

MILITARY SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, MULTICONDUCTOR, (3 CONDUCTORS) AND SINGLE SHIELDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and and Standards (DODISS) specified in the solicitation: MIL-C-27072.

Inactive for new design. For new design use MIL-C-55021/2.

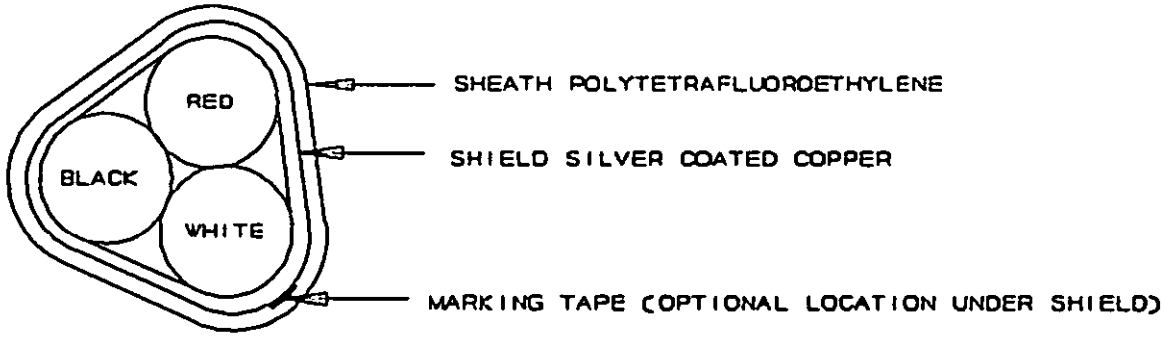


FIGURE 1. Cable configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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DESIGN DATA:

Design and construction: See figure 1.

Number of conductors: Three.

Conductor size and stranding: See table I.

Wire: In accordance with M16878/4, or /21 (see table I for complete Part or Identifying Number (PIN)).

Jacket undershield: Not applicable.

Marking tape: Marking tape shall be placed under the sheath or shield.

Shielding: Silver coated copper.

Sheath: Polytetrafluoroethylene.

Temperature rating: 200°C.

REQUIREMENTS:

Visual and dimensional:

Components:

Basic wire: Silver-coated copper.

Dimension: See table I.

Primary insulation:

Material: Polytetrafluoroethylene.

Colors:

White.

Black.

Red.

Spark test voltage: 3,400 V rms.

Inspection test voltage: 2,000 V rms.

Impulse dielectric test: 6,500 V peak.

Shield:

Material: Silver coated copper

Construction: See table I.

Braid angle: 20° to 40°.

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Workmanship: In accordance with MIL-C-27072.

Sheath:

Material: Polytetrafluoroethylene.

Wall thickness: .011 inch ±.003.

Tensile strength: In accordance with table V of MIL-C-27072.

Tension set: Not applicable.

Tear strength: Not applicable.

Accelerated aging: Not applicable.

Oil resistance: Not applicable.

Surface resistance (sheath): When tested in accordance with method 6041 of FED-STD-228, the initial surface resistance shall be not less than 5 megohms. The final surface resistance shall be within 50 percent of the initial surface resistance. (NOTE: The specimen for this inspection test shall not be subjected to any other electrical tests.)

Color: See table IV of MIL-C-27072.

Finished cable:

Diameter: In accordance with table I.

Conductor resistance: In accordance with MIL-W-16878/4 or MIL-W-16878/21.

Dielectric strength: In accordance with MIL-W-16878/4 or MIL-W-16878/21.

Cold bend: Mandrel size in accordance with table XIII of MIL-C-27072.

Heat resistance: A specimen of finished cable shall be subjected to 250°C ±5°C in an air oven for 96 hours minimum. It shall be removed from the oven and allowed to cool to room temperature. After 1 hour at room temperature, the insulation and jacket shall be checked for shrinkage of not more than .125 inch. The specimen shall then be wound 5 close turns around the mandrel of 10 times the cable diameter. In this position, the specimen shall be subjected to the dielectric strength test. If the cable fails the dielectric strength test or shrinks more than .125 inch, the lot shall be rejected.

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TABLE I. Dimensions.

PIN M27072/89	Conductor			Nom. cable lay	Overall shield					Finished cable dia. (in)		Approx weight per 1000 ft (lb)	Basic wire M16878/4 M16878/21
	AWG	Stranding (AWG)	Nom. dia. (in.)		Parameters 1/								
					d	C	N	P	D	Min.B	Max.		
TDDE*	28	7/36	.015	1.00	.005	18	4	18	.084	.101	.113	12.9	BCB***
SDDE*	26	7/34	.0189	1.10	.005	18	5	17	.093	.110	.126	16.2	BDB***
NDDE*	24	19/36	.0248	1.25	.005	18	5	16	.103	.119	.137	18.9	BEE***
LDDE*	22	19/34	.0312	1.40	.005	16	5	15	.116	.133	.149	23.2	BFE***
JDDE*	20	19/32	.0396	1.60	.005	16	6	13	.133	.150	.166	29.7	BGE***
GDDE*	18	19/30	.0495	1.90	.005	16	8	12	.159	.171	.189	39.9	BHE***
FDDE*	16	19/29	.057	2.20	.005	16	7	10	.187	.188	.212	48.6	BJE***

1/ See MIL-C-27072, overall shielding. Optional shields are acceptable that meet the requirements of MIL-C-27072.

Asterisks (*) shall be replaced with the appropriate color of the cable sheath in accordance with MIL-STD-681, table I. If a strip or a band is to be added it shall be in accordance with 3.2.2 of MIL-C-27072.

Revision changes are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
DLA - ES, IS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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